Jinchen Zhang

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EDUCATION

Tsinghua University **Beijing, China**
B.S. in Automation, GPA: 3.3/4.0 09/2023 - 06/2027 (expected)
Relevant Courses: Calculus, Linear Algebra, Random Mathematics and Statistics, Discrete Mathematics, Operations Research, Computer Languages and Programming, Data Structures, Object-Oriented Programming Training, Computer Networks and Applications, Pattern Recognition and Machine Learning, Intelligent Speech Processing, AI for Science, Intelligent Measurement System

PUBLICATIONS AND PREPRINTS

[1] *Strength Controls Change How Interventions Compare Across Sparse Autoencoder Codes.*
Jinchen Zhang*, Zheyuan Liu*, Yihan Zhu, and Meng Jiang. 2026.
arXiv:2610.26489. Under review at ACL Rolling Review (ARR). **Equal contribution.*

[2] *Training Negatives Still Matter After Calibrating Answerability Readouts.*
Jinchen Zhang, Zheyuan Liu, Yihan Zhu, and Meng Jiang. 2026.
arXiv:2610.26461. Under review at ARR.

RESEARCH EXPERIENCE

Trustworthy LLMs: Interpretability and Answerability **Notre Dame, IN**
Advised by Prof. Meng Jiang, DM2 Lab, University of Notre Dame 07/2026 - Present

- Led research, experiments, and manuscript writing for two first/co-first-author papers [1, 2].
- **SAE intervention evaluation [1]:** Compared sparse autoencoder interventions across five experimental setups involving Gemma and Llama models.
- On 304 new Llama inputs, full-output KL calibration preserved all reference transfer decisions; total-norm matching lost 11 of 175 reference transfers.
- **Answerability estimation [2]:** Evaluated predictors of whether a passage supports an answer using Qwen and Gemma on SQuAD 2.0 and RepLiQA.
- For selected logistic readouts, retaining versus removing negative passages improved calibrated SQuAD 2.0 AUROC from 0.923 to 0.947 (Qwen) and 0.904 to 0.934 (Gemma).

Visual Instruction Pretraining (ViTP) for Human Sensing **Remote**
Advised by Prof. Yuchen Liu, North Carolina State University 01/2026 - 02/2026

- Constructed 1,946 MMFi training examples linking depth frames, action descriptions, and skeleton cues; applied sequence-level splits and filtered corrupted frames.
- Adapted ViTP with InternVL2.5 for action recognition; fine-tuned vision and projection layers with a frozen LLM and implemented multi-frame evaluation.

Efficient Sparse Modeling for Vision-Language Models **Remote**
Advised by Prof. Yinglun Zhu, University of California, Riverside 07/2025 - 10/2025

- Independently proposed a structured-sparsity approach for efficient multimodal learning, designed the architecture, implemented a prototype, and tested it on synthetic data.

Deep Learning-Based Vehicle Trajectory Prediction **Beijing, China**
Advised by Prof. Li Li, Tsinghua University 09/2024 - 09/2025

- Developed a deep learning framework for vehicle trajectory prediction, targeting accuracy and real-time performance in dynamic traffic environments.
- Built a visualization system displaying predicted trajectories alongside road layouts and vehicle states.
- Supported BEVFormer comparisons by standardizing experimental settings and automating run logs.

Development of S&P 500 Quantitative Trading Strategy **Remote**
Advised by Prof. Vipul Goyal, Carnegie Mellon University 08/2024 - 12/2024

- Developed a PyTorch framework to identify S&P 500 trading signals and support automated decisions.
- Compared the model with baseline portfolio strategies to assess robustness.

SKILLS

Languages: Chinese (native); English (proficient); Spanish (basic communication)
Technical Skills: Python, PyTorch, C, C++